Customers

1. **Who are the high-valued customers?**

SQL query:

```
WITH customer_spending AS (
  SELECT
    customerid,
    SUM(contractprice) AS total_spending
  FROM engagements
  GROUP BY customerid
),
segmented AS (
  SELECT
    customerid,
    total_spending,
    NTILE(3) OVER (ORDER BY total_spending DESC) AS segment
  FROM customer_spending
)
SELECT *
FROM segmented;
```

customerid	total_spending	segment
10005	25585	1
10006	13170	1
10014	12770	1
10010	12455	1
10002	12320	1
10004	11800	2
10001	10795	2
10015	8255	2
10013	7560	2
10003	7250	3
10009	7090	3
10012	6815	3
10007	4685	3

123 customerid	123 total_spending	123 segment
10,005	25,585	1
10,006	13,170	1
10,014	12,770	1
10,010	12,455	1
10,002	12,320	1
10,004	11,800	2
10,001	10,795	2
10,015	8,255	2
10,013	7,560	2
10,003	7,250	3
10,009	7,090	3
10,012	6,815	3
10,007	4,685	3

TuneWorks is an entertainment agency, and customer value directly impacts its revenue structure and long-term profitability. Therefore, identifying which customers contribute the most spending is a critical first step. By calculating total spending and using NTILE(3) to segment customers (1= high-value, 2 = mid-value, 3 = low-value), we observed that a small group of customers contributes a disproportionately large share of total spending, indicating strong revenue concentration. Customers 10008 and 10011 did not appear in the segmentation output, likely because they have no recorded engagements and therefore no spending.

Once the high-value customers are identified, the next logical step is to understand what drives their spending—specifically, the musical styles and entertainer types they prefer. These insights will guide the following analyses and help TuneWorks determine whether its top customers share similar tastes, whether current offerings align with those preferences, and how the company should allocate resources to better retain its most valuable clients.

2. Musical Style Preferences of High-Value Customers

SQL query:

```
WITH customer_spending AS (
  SELECT
    customerid,
    SUM(contractprice) AS total_spending
  FROM engagements
  GROUP BY customerid
),
segmented AS (
  SELECT
    customerid,
    total_spending,
```

```
    NTILE(3) OVER (ORDER BY total_spending DESC) AS segment
  FROM customer_spending
),
high_value_customers AS (
  SELECT customerid
  FROM segmented
  WHERE segment = 1
)
SELECT
  ms.StyleName,
  COUNT(*) AS preference_count
FROM engagements e
JOIN high_value_customers hvc
  ON e.customerid = hvc.customerid
JOIN Entertainer_Styles es
  ON e.entertainerid = es.entertainerid
JOIN Musical_Styles ms
  ON es.StyleID = ms.StyleID
GROUP BY ms.StyleName
ORDER BY preference_count DESC;
```

stylename	preference_count
60's Music	15
Standards	14
Country	13
Contemporary	11
Top 40 Hits	11
Show Tunes	10
70's Music	7
Classic Rock & Roll	7
Country Rock	5
Variety	5
Salsa	5
Jazz	5
Classical	4
Chamber Music	3
Folk	3

Rhythm and Blues	2
Motown	1

A-Z stylename	123 preference_count
60's Music	15
Standards	14
Country	13
Contemporary	11
Top 40 Hits	11
Show Tunes	10
70's Music	7
Classic Rock & Roll	7
Country Rock	5
Variety	5
Salsa	5
Jazz	5
Classical	4
Chamber Music	3
Folk	3
Rhythm and Blues	2
Motown	1

After identifying the high-value customers, we examined the musical styles they book most often. By filtering engagements to include only segment-1 customers, the analysis reveals that 60's Music, Standards, Country, and Contemporary dominate their preferences, while niche styles such as Motown, Folk, and Chamber Music appear only rarely. This pattern suggests that high-value customers tend to favor familiar and broadly appealing genres. From a business perspective, this indicates that TuneWorks should ensure strong entertainer coverage in the top styles to better serve and retain its most profitable clients. These findings naturally lead to the next question: within these preferred styles, which entertainer do high-value customers choose most frequently?

3. Entertainer Preferences of High-Value Customers

SQL query:

```
WITH customer_spending AS (
  SELECT
    customerid,
    SUM(contractprice) AS total_spending
  FROM engagements
```

```
  GROUP BY customerid
),
segmented AS (
  SELECT
    customerid,
    total_spending,
    NTILE(3) OVER (ORDER BY total_spending DESC) AS segment
  FROM customer_spending
),
high_value_customers AS (
  SELECT customerid
  FROM segmented
  WHERE segment = 1
)
SELECT
  en.EntStageName AS EntertainerName,
  COUNT(*) AS booking_count
FROM engagements e
JOIN high_value_customers hvc
  ON e.customerid = hvc.customerid
JOIN Entertainers en
  ON e.entertainerid = en.entertainerid
GROUP BY en.EntStageName
ORDER BY booking_count DESC;
```

entertainer_name	style_name	booking_count
Country Feeling	60's Music	8
Country Feeling	Country	8
JV & the Deep Six	Classic Rock & Roll	7
Carol Peacock Trio	Standards	7
Saturday Revue	Standards	7
JV & the Deep Six	60's Music	7
Carol Peacock Trio	Contemporary	7
Saturday Revue	70's Music	7
Carol Peacock Trio	Show Tunes	7
Saturday Revue	Top 40 Hits	7
Coldwater Cattle Company	Country Rock	5
Coldwater Cattle Company	Country	5
Modern Dance	Variety	4
Caroline Coie Cuartet	Contemporary	4
Caroline Coie Cuartet	Jazz	4
Modern Dance	Salsa	4
Modern Dance	Top 40 Hits	4
Julia Schnebly	Show Tunes	3
Julia Schnebly	Chamber Music	3
Julia Schnebly	Classical	3
Jim Glynn	Folk	2
Topazz	Rhythm and Blues	1
Topazz	Motown	1
Susan McLain	Folk	1
Jazz Persuasion	Jazz	1
Jazz Persuasion	Rhythm and Blues	1
Susan McLain	Classical	1
Jazz Persuasion	Salsa	1
Topazz	Variety	1

To verify whether the musical style preferences of high-value customers translate into actual booking behavior, we linked each entertainer with their corresponding musical style and analyzed the number of bookings. The results clearly reinforce previous findings: the artists booked most frequently by high-value customers, such as Country Feeling, JV & the Deep Six, the Carol Peacock Trio, and Saturday Revue. They all belonged to the previously identified most popular music genres.

Importantly, the data shows that high-value customers not only exhibit a highly concentrated preference for musical styles, but the artists they choose also closely align with these styles. This indicates that their preferences are stable and pattern-driven, rather than random or highly variable.

In contrast, artists associated with low-preference styles (such as ballroom, chamber music, and folk) appeared only once, further confirming this idea.

Overall, this adjustment has clear implications for TuneWorks' business development: prioritizing and promoting artists of high-demand types, as they are crucial for retaining and maximizing the value of the company's most profitable clients.

4. Booking Frequency Trend of High-Value Customers Over Time

SQL query:

```
WITH customer_spending AS (
  SELECT
    customerid,
    SUM(contractprice) AS total_spending
  FROM engagements
  GROUP BY customerid
),
segmented AS (
  SELECT
    customerid,
    total_spending,
    NTILE(3) OVER (ORDER BY total_spending DESC) AS segment
  FROM customer_spending
),
high_value_customers AS (
  SELECT customerid
  FROM segmented
  WHERE segment = 1
)
SELECT
  e.customerid,
  EXTRACT(YEAR FROM e.startdate) AS year,
  COUNT(*) AS num_bookings
FROM engagements e
JOIN high_value_customers hvc
  ON e.customerid = hvc.customerid
GROUP BY
  e.customerid,
  EXTRACT(YEAR FROM e.startdate)
ORDER BY
  e.customerid,
  year;
```

123 customerid	123 year	123 num_bookings
10,002	2,017	7
10,002	2,018	3
10,005	2,017	6
10,005	2,018	2
10,006	2,017	6
10,006	2,018	3
10,010	2,017	7
10,010	2,018	6
10,014	2,017	5
10,014	2,018	5

Looking at the year-to-year booking counts for high-value customers, we can see that their activity levels are not consistent. Some customers (such as 10002, 10005, and 10006) booked noticeably less in 2018 compared to 2017, while others (10010 and 10014) maintained a steady level of engagement. This mixed pattern shows that even within the top customer segment, behavior varies, and not all high-value customers remain equally active over time.

Although the data does not explain why certain customers decreased their bookings, this simple trend analysis still provides useful signals. It helps us identify which high-value customers may be at risk of becoming less engaged and which ones remain stable. For a business like TuneWorks—where revenue is concentrated among a small group of clients—keeping an eye on these changes is especially important. Tracking booking frequency over time gives the company an early indication of where retention efforts may be needed and which relationships are likely to stay strong.

5. **What is their booking frequency?**
6. **Are high-valued customers increasing or decreasing their booking frequency over time?**

7. segment our customers

1️⃣ **Who are our high-valued customers?**

2️⃣ **What are their key preferences in musical styles and entertainer types?**

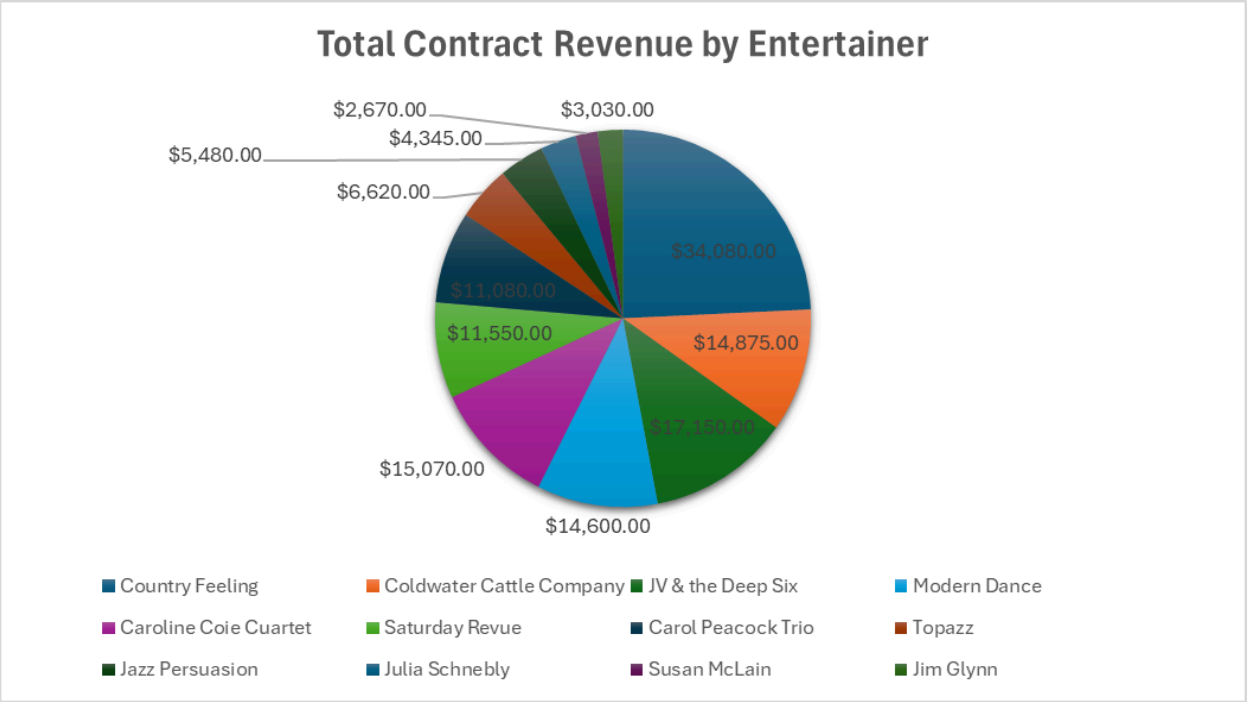
3️⃣ **What does their booking behavior look like, including frequency and trends over time?**

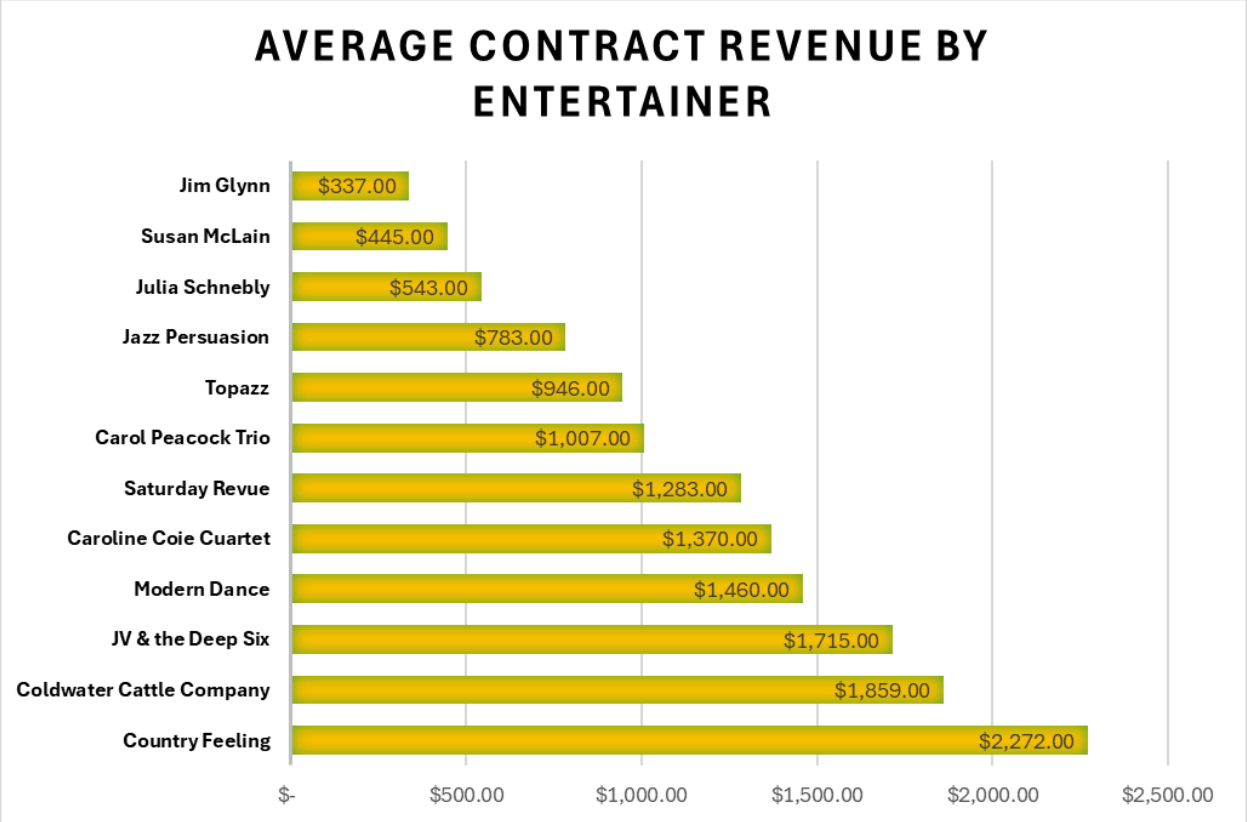
4️⃣ **How can we segment customers based on value, preferences, and behavior?**

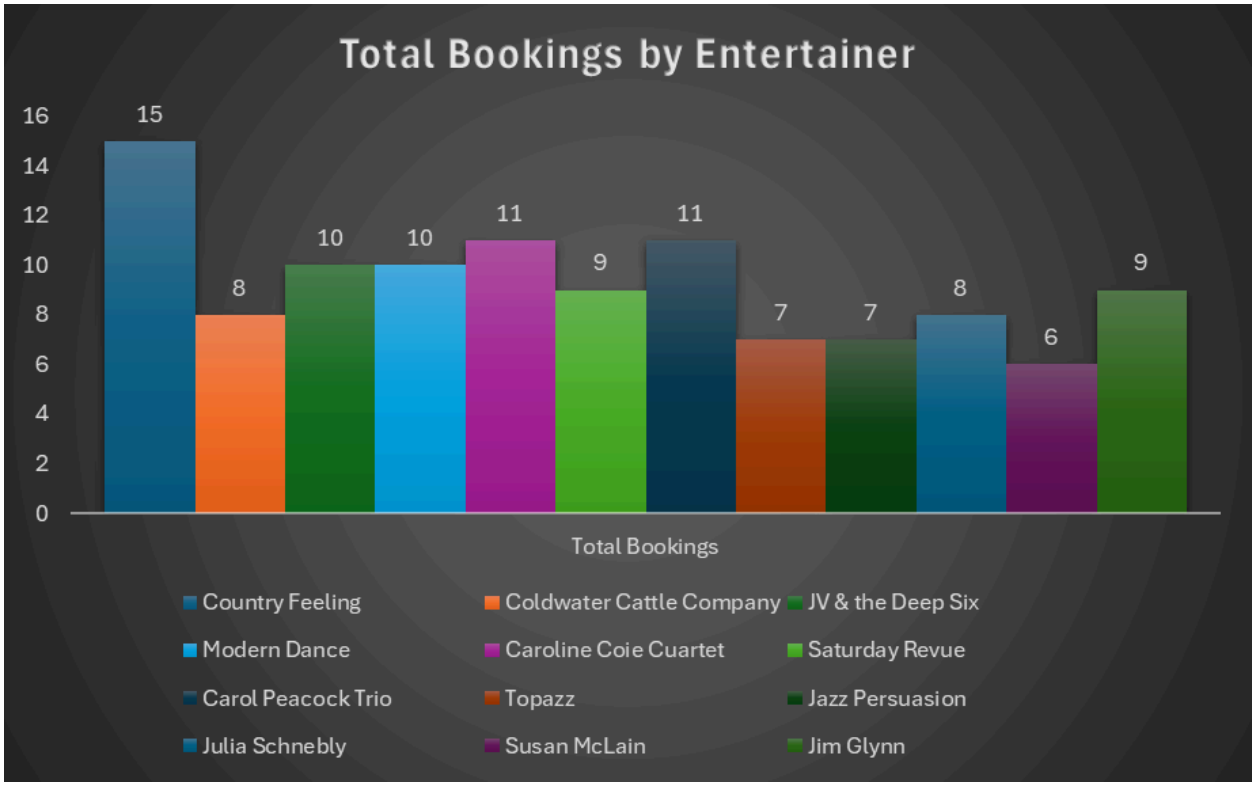
Data Visualizations

Query 1

Query 1: Total Revenue by Entertainer, Average Revenue by Entertainer, Total Bookings

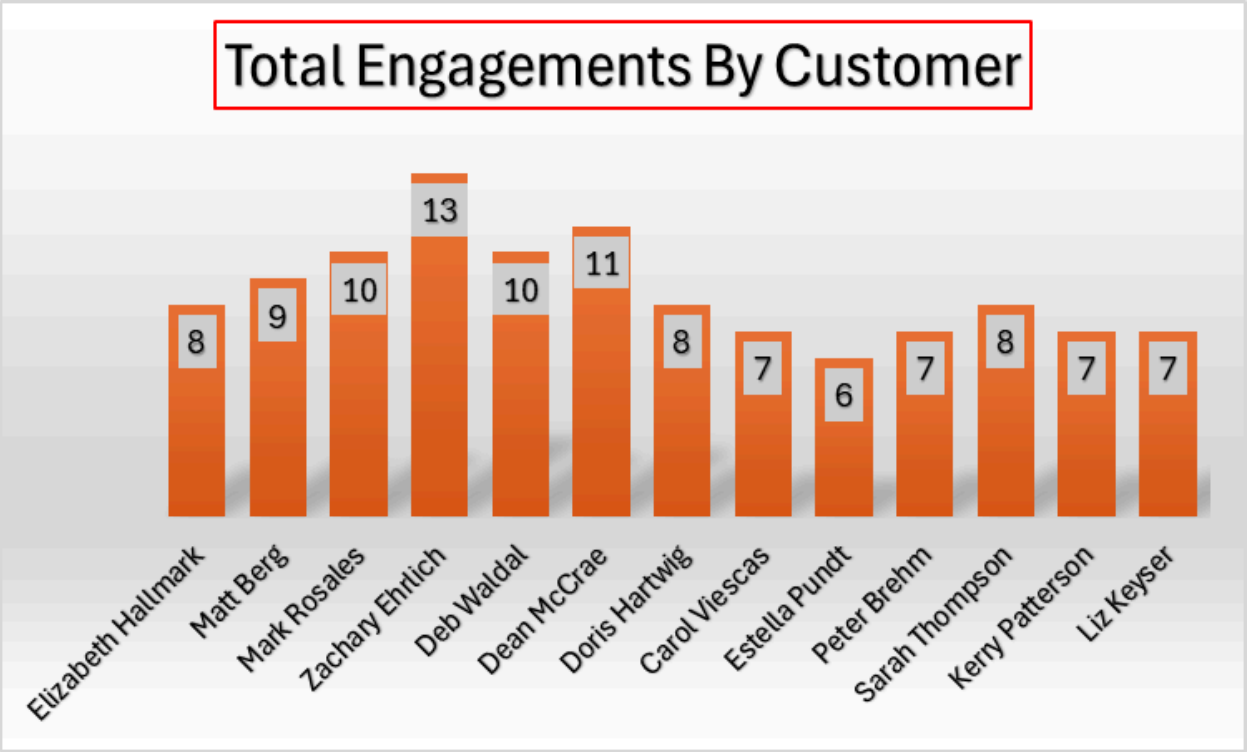


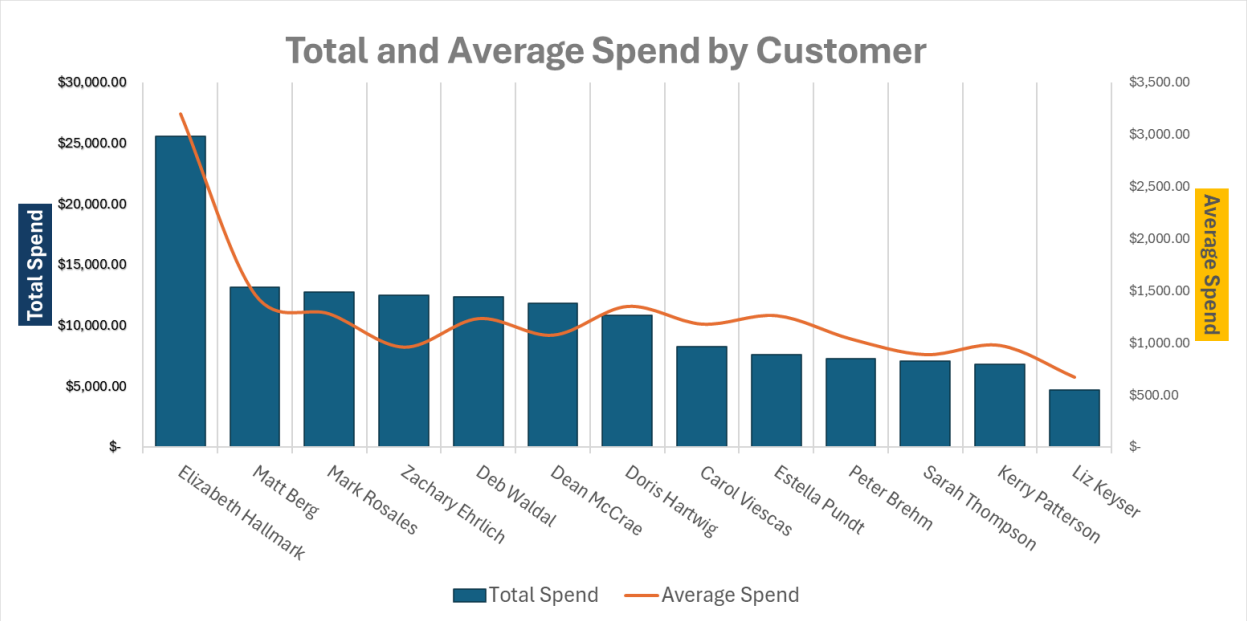



Total Bookings by Entertainer
16
14
12
10
8
6
4
2
0
15
8
10
10
11
9
11
7
7
8
6
9
Total Bookings
Country Feeling
Coldwater Cattle Company
JV & the Deep Six
Modern Dance
Caroline Coie Cuartet
Saturday Revue
Carol Peacock Trio
Topazz
Jazz Persuasion
Julia Schnebly
Susan McLain
Jim Glynn

Query 2

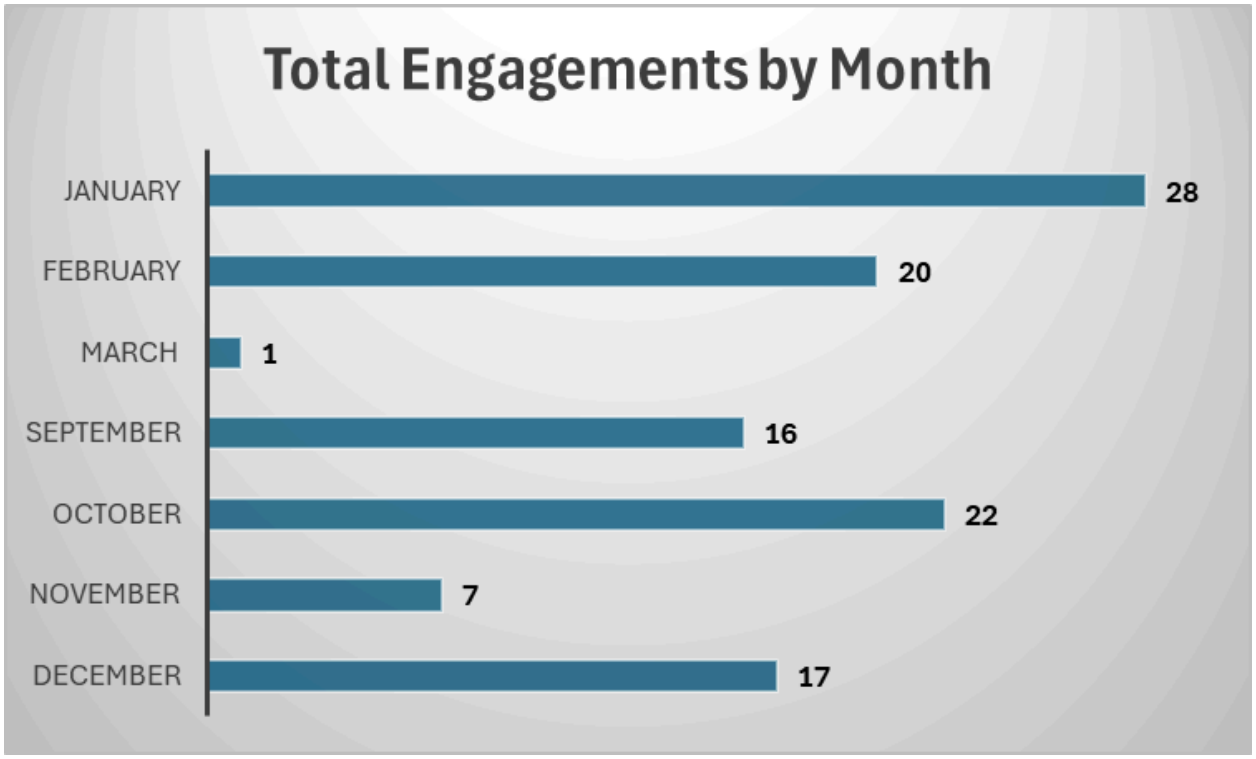
Query 2: Customer Analysis: Total Engagements, Total Spend, Average Spend





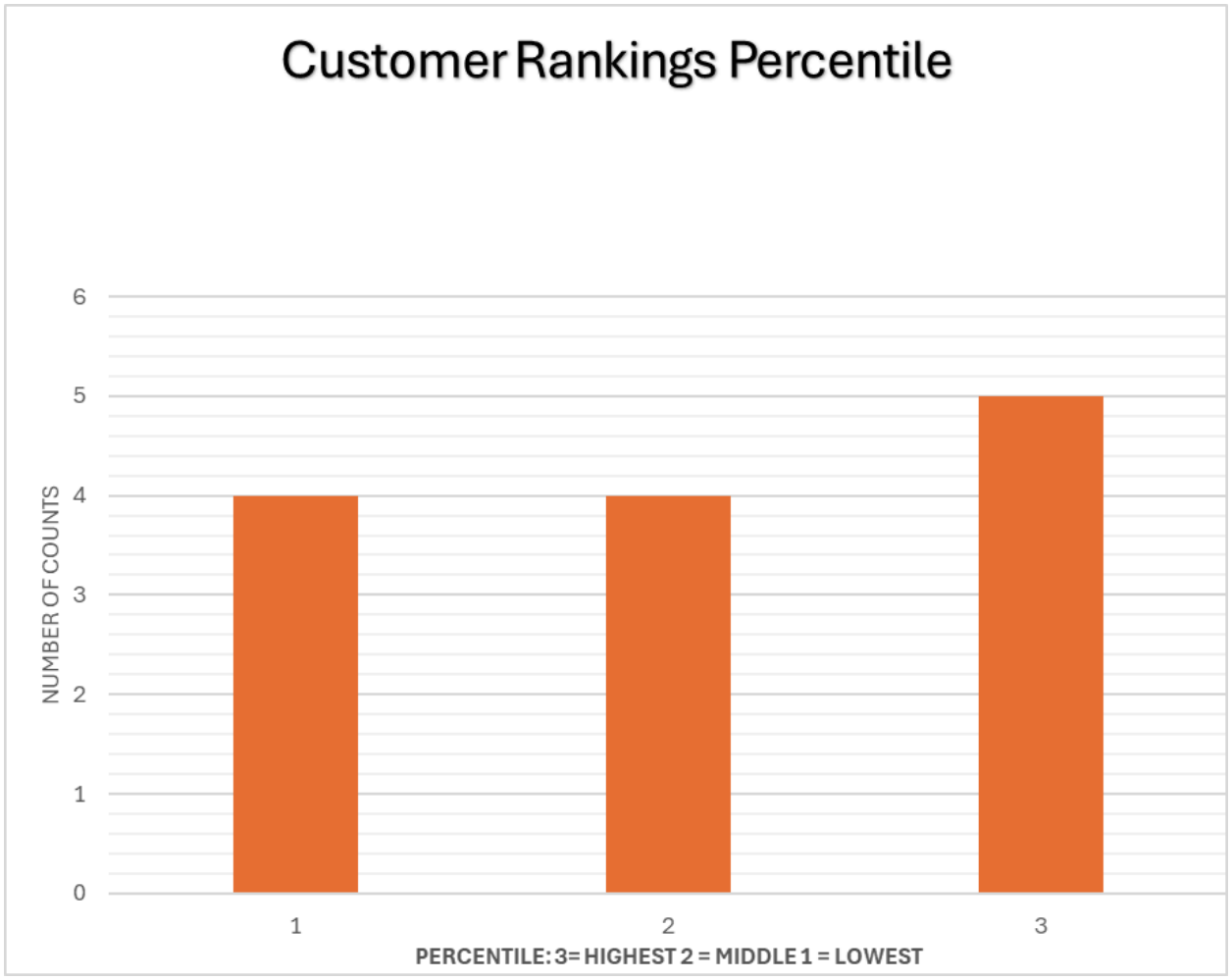
Query 3

Query 3 - which months have the most engagement?

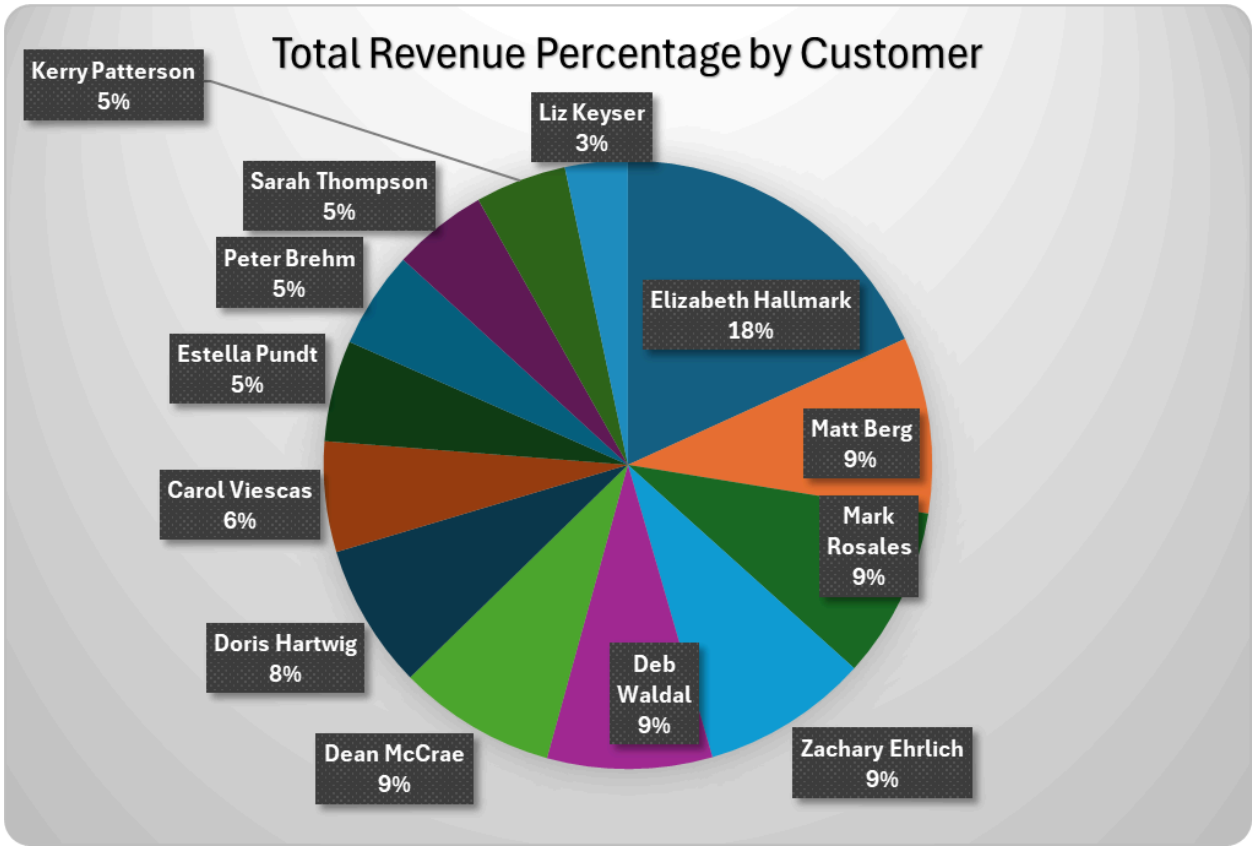


Query 4

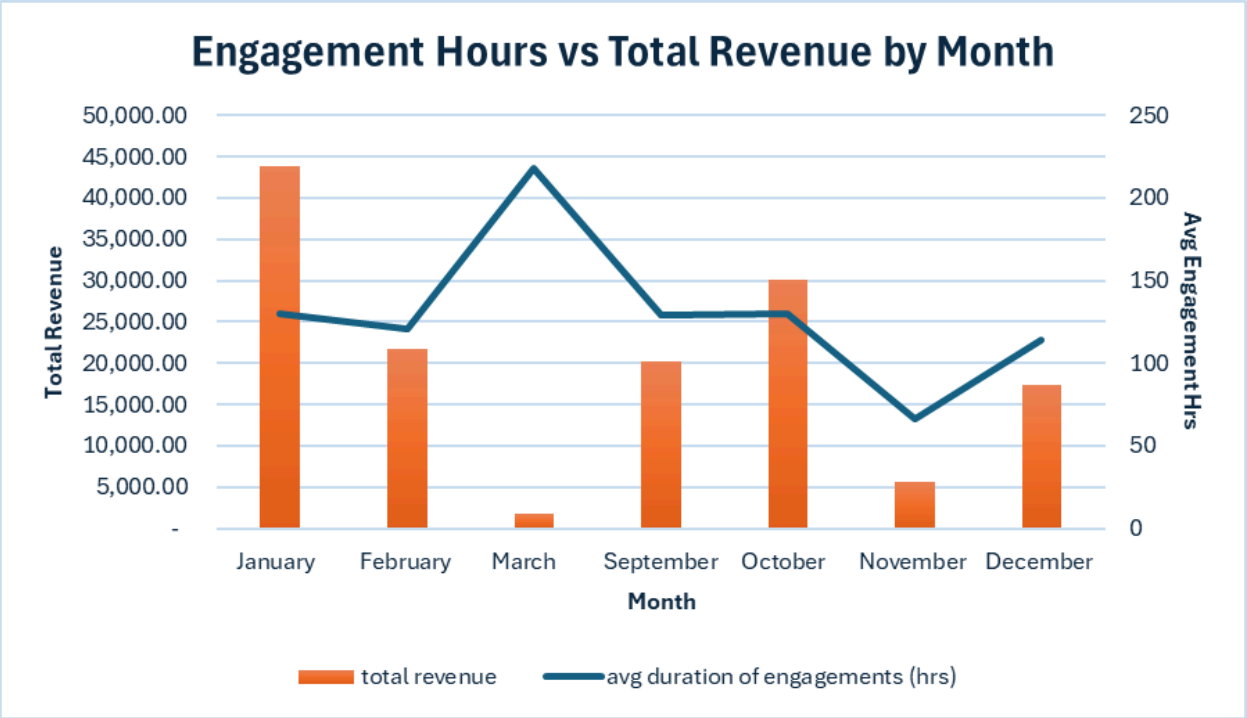
Overlaps with query 1



Query 5

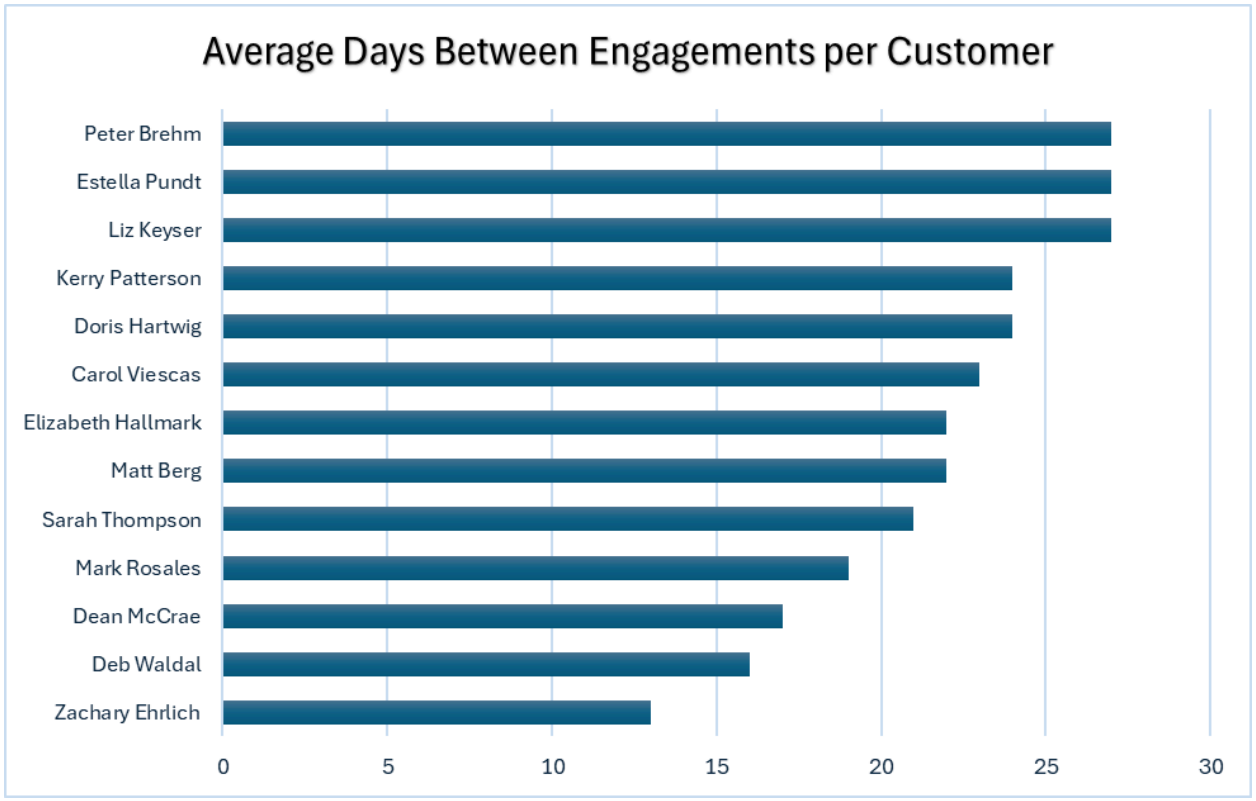

Total Revenue Percentage by Customer
Kerry Patterson 5%
Liz Keyser 3%
Sarah Thompson 5%
Peter Brehm 5%
Elizabeth Hallmark 18%
Estella Pundt 5%
Matt Berg 9%
Carol Viescas 6%
Mark Rosales 9%
Doris Hartwig 8%
Deb Waldal 9%
Dean McCrae 9%
Zachary Ehrlich 9%

Query 6



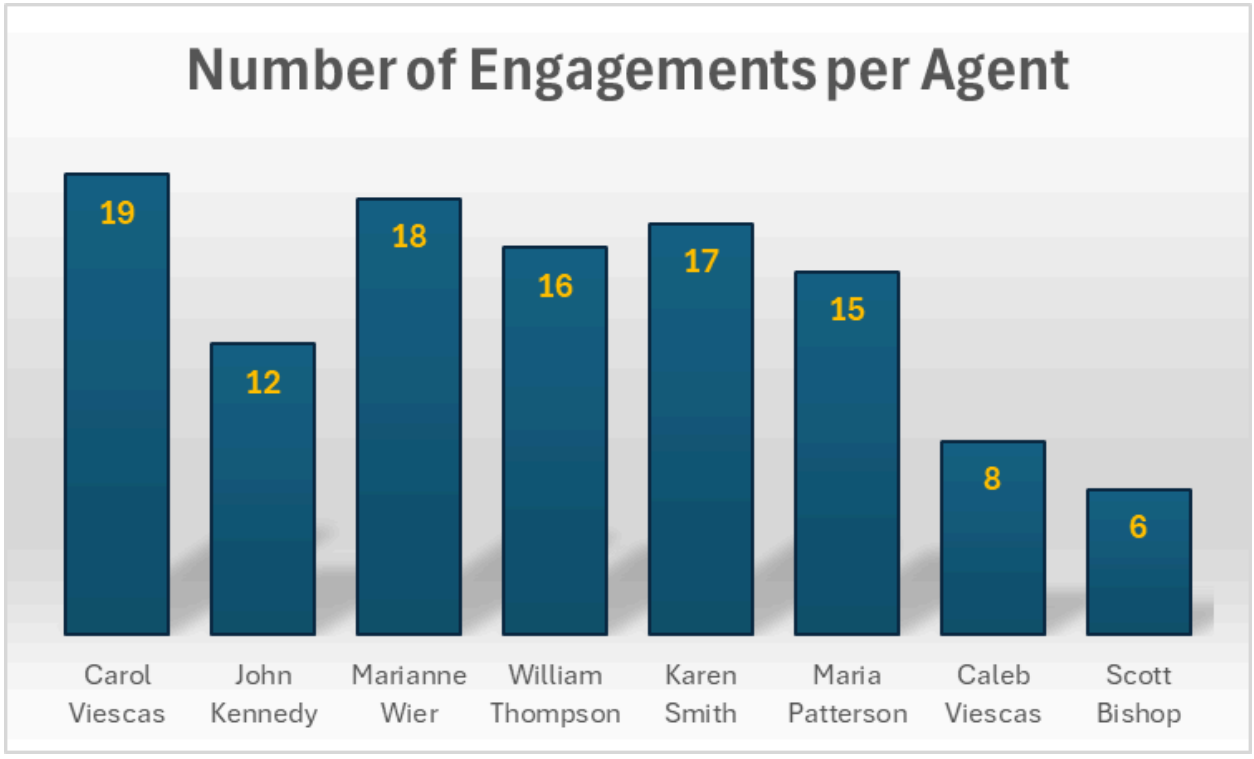
****Combine in slide with Query 3**

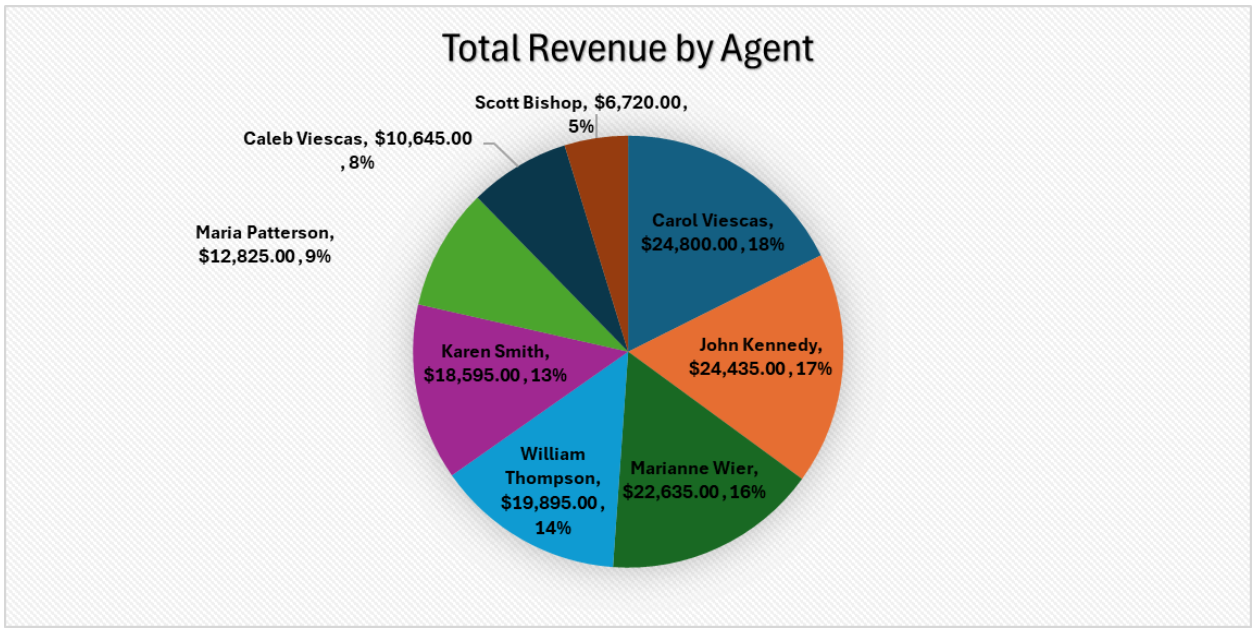
Query 7


Average Days Between Engagements per Customer
Peter Brehm
Estella Pundt
Liz Keyser
Kerry Patterson
Doris Hartwig
Carol Viescas
Elizabeth Hallmark
Matt Berg
Sarah Thompson
Mark Rosales
Dean McCrae
Deb Waldal
Zachary Ehrlich
0
5
10
15
20
25
30

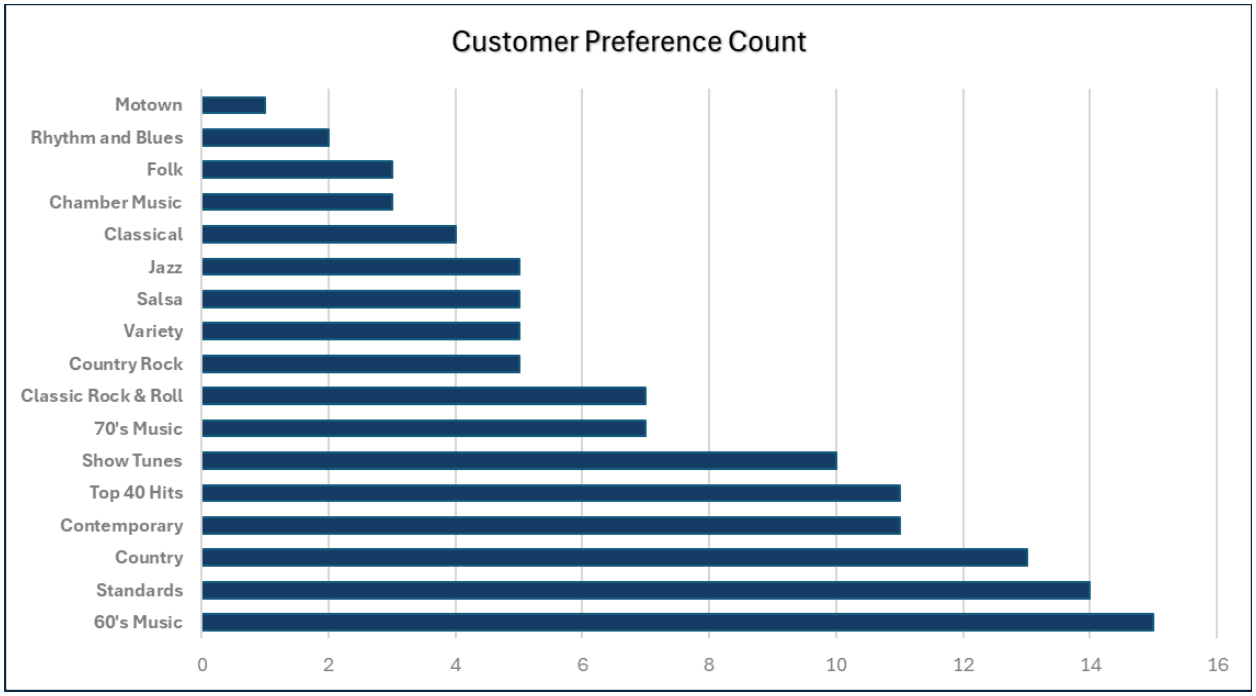
Query 8

Agent Analysis

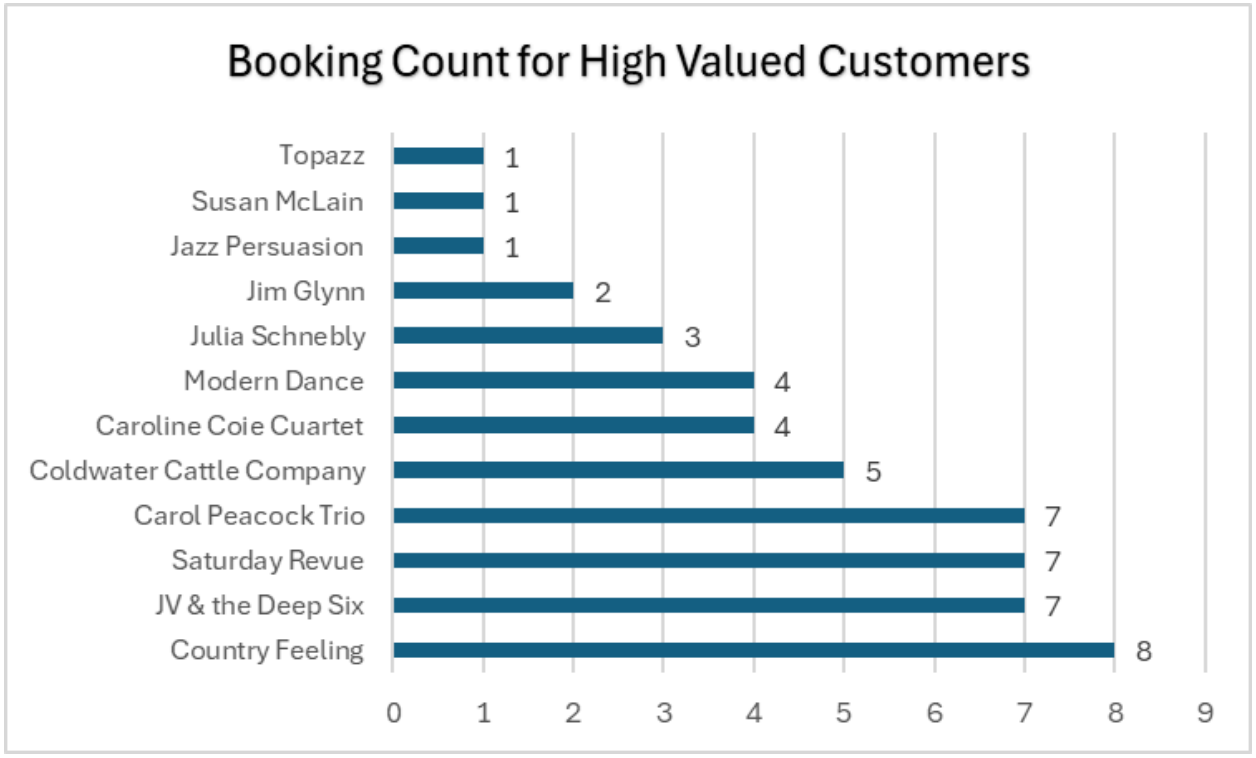




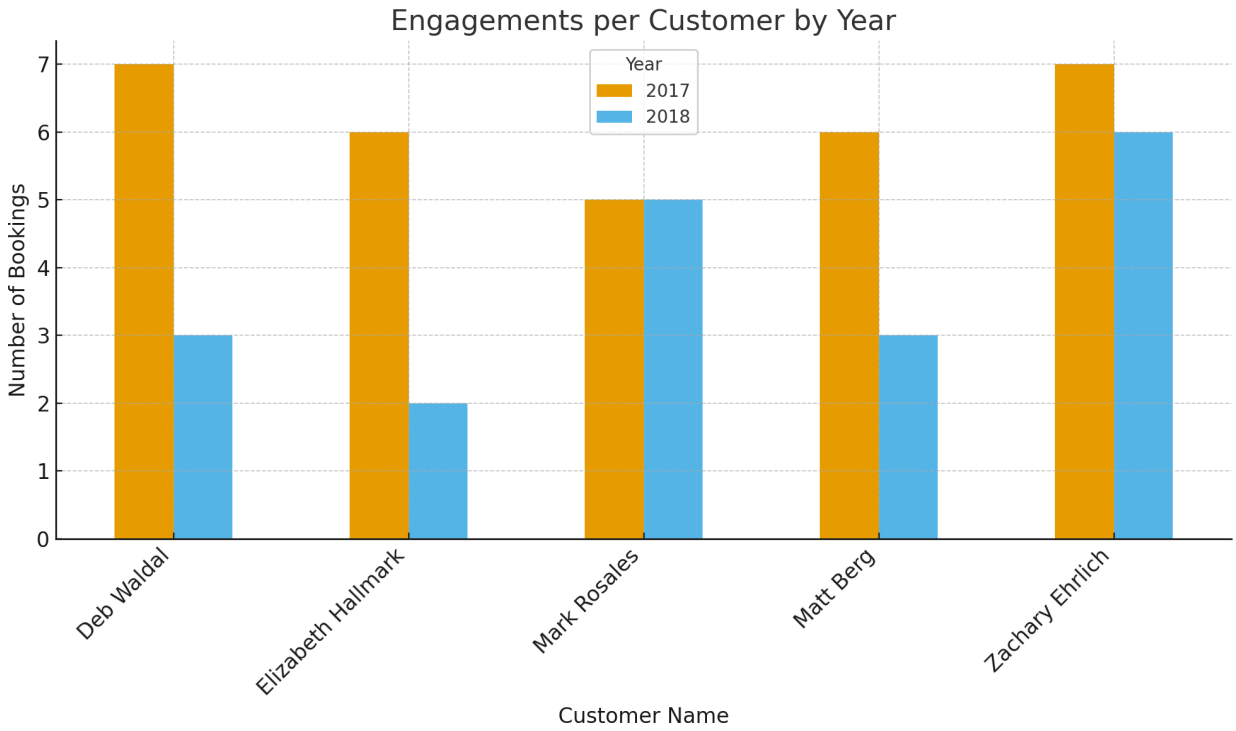
Query 9


Customer Preference Count
Motown
Rhythm and Blues
Folk
Chamber Music
Classical
Jazz
Salsa
Variety
Country Rock
Classic Rock & Roll
70's Music
Show Tunes
Top 40 Hits
Contemporary
Country
Standards
60's Music
0
2
4
6
8
10
12
14
16

Query 10


Booking Count for High Valued Customers
Topazz
Susan McLain
Jazz Persuasion
Jim Glynn
Julia Schnebly
Modern Dance
Caroline Coie Cuartet
Coldwater Cattle Company
Carol Peacock Trio
Saturday Revue
JV & the Deep Six
Country Feeling
1
1
1
2
3
4
4
5
7
7
7
8
0
1
2
3
4
5
6
7
8
9

Query 11


Engagements per Customer by Year
Year
2017
2018
Number of Bookings
0
1
2
3
4
5
6
7
Deb Waldal
Elizabeth Hallmark
Mark Rosales
Matt Berg
Zachary Ehrlich
Customer Name